A6 · Who Carries Your Packets?

CS425 Computer Networks · instructor led · every command runs on Onyx (`ssh onyx`)

TURN THIS IN ON PAPER
before you leave the room
graded pass/fail

Name _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · WHO OWNS ONYX'S ADDRESS? `dig +short TXT ... .cymru.com`

QUESTION	ANSWER
Predict: prefix length Boise State announces for Onyx's address (Onyx's own table says /25)	
AS number and name · prefix announced	
How many addresses are in that prefix?	

Q1. Why announce one big prefix instead of all the little subnets? Who knows about the /25, and which kind of routing protocol tells them?

ROUND 2 · MAP THE TRACE ONTO THE ASES `mtr -n -z -T -P 443 -r -c 3 mirrors.xmission.com`

QUESTION	ANSWER
Predict: how many ASes do the 12 routers belong to?	
The AS path from Onyx to XMission, as numbers	
The same path, as company names	

Q2. Between which pairs of hops does the packet cross from one AS into another (where eBGP lives)? Which hops are inside one AS, where an intra-AS protocol like OSPF decides?

ROUND 3 · BOISE STATE FROM THE OUTSIDE `RIPEstat: asn-neighbours, looking-glass`

QUESTION	ANSWER
Predict: how many ASes does Boise State exchange routes with?	
Which ones? (number and name)	
Predict: of 298 paths, how many through Zayo · through IRON?	
Actual: through Zayo · through IRON	

Q3. Why would a university pay for two providers instead of one? Both neighbors are `left` and none are `right`: does Boise State carry traffic **between** other networks? If AS 6461 got a route whose AS-PATH already contained 6461, what would it do, and why?

ROUND 4 · NEAR IS NOT FAST `mtr -n -z -T -P 443 -r -c 3 8.8.8.8`

QUESTION	ANSWER
Predict: faster to answer, Google's 8.8.8.8 or XMission (475 km)?	
Predict: ASes on the way to Google	
Actual: time to Google · time to XMission · AS path to Google	

Q4. Light in fiber covers about 200 km per millisecond. How far away could the 8.8.8.8 that answered be? Google announces 8.8.8.8 from many places (anycast): which copy do you reach, and who decides? The first Lumen router on the XMission path is already 27 ms away. What does that suggest?

Exit. Boise State wants all outbound traffic to go through IRON whenever IRON can reach the destination, even when the path through Zayo is shorter. Which BGP route selection rule does that, and why does the shorter AS-PATH not win?

Graded pass/fail on something in every box. Hand this in before you leave.